One-One & Onto Linear Transformation

Step-I: Matrix of the transformation

Step-II: Reduce the matrix into Echelon form

Step-III: Identify the pivot Rows & Columns.

If each column is the pivot columns (One-One) ⇓ Unique sol.

If each row is the pivot row (onto) ⇓ Always consistent.

$T: \mathbb{R}^2 \to \mathbb{R}^3$

$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ x-y \\ 0 \end{pmatrix}$. ✓

$A = \begin{pmatrix} 2 & 0 \\ 1 & -1 \\ 0 & 0 \end{pmatrix} \xrightarrow{R} \begin{pmatrix} 1 & -1 \\ 2 & 0 \\ 0 & 0 \end{pmatrix} \xrightarrow{R} \begin{pmatrix} 1 & -1 \\ 0 & 2 \\ 0 & 0 \end{pmatrix}$

All the columns → pivot ⇒ One-One.

Third Row doesn't have pivot ⇒ Not Onto.

$T: \mathbb{R}^2 \to P_1$

$T\begin{pmatrix} a \\ b \end{pmatrix} = a + (a+b)x$

$T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 + x$ ✓

$T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = x$ ✓

$[1+x]_C = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$[x]_C = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$.

R $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

T is one-one & Onto.

1-1 $\Leftrightarrow$ Onto

* $A_{n \times n} \longrightarrow$

$T: P_2 \to \mathbb{R}^2$

$T(a+bx+cx^2) = \begin{pmatrix} a-b \\ b+c \end{pmatrix}$

$B = \{1, x, x^2\}$

$C = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$

$T(x) = T(0+1x+0x^2) = \begin{pmatrix} 0-1 \\ 1+0 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

Step-1:

$T(1) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$T(x) = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$T(x^2) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\Rightarrow A = \begin{pmatrix} \triangle & -1 & 0 \\ 0 & \triangle & 1 \end{pmatrix}$

T is onto

Non-pivot column

$\Rightarrow$ T is NOT one-one

$T: V \to W$

$\dim(V) > \dim(W)$ | never 1-1

No. of cols > No. of Rows

$\dim(V) < \dim(W)$

No. of cols < No. of Rows

Never onto

Inverse of linear transformation

T → 1-1 and onto → Bijective.

Bijective ↓

Invertible.

$T: \mathbb{R}^2 \to \mathbb{R}^2$

$T(x) = Ax$

$T^{-1}(x) = A^{-1}x$.

$T: \mathbb{R}^2 \to \mathbb{R}^3$ OR $T: \mathbb{R}^3 \to \mathbb{R}^2$

$T: V \to W$

$\dim(V) \neq \dim W$

$\Rightarrow$ Either Not 1-1 $\Rightarrow$ Not OR Not onto Bijective